MATH11219

LECTURE NOTES

Authored by
Lachlan J. W. Wyer

Document Start Date: 27th of Jan 2026
Latest Update: January 28, 2026

Contents

1 Differentiation **2**
1.1 Average Rate-of-Change (ARoC): 2
1.2 Instaneous Rate-of-Change (IRoC): 2
1.3 Limits in Functions: 2
1.4 Common Derivatives: 4

1 Differentiation

Differentiation (Derivatives) is just a mathematical expression of a functions rate-of-change. Think of the gradient of a function, area/estimation of a function, kinematics & optimisation of area or numbers.

1.1 Average Rate-of-Change (ARoC):

The ARoC of the function is just the estimation of a curve based on a linear lines gradient. I.E. the average gradient between two points.

$$\text{ARoC} = \frac{y_2 - y_1}{x_2 - x_1}$$

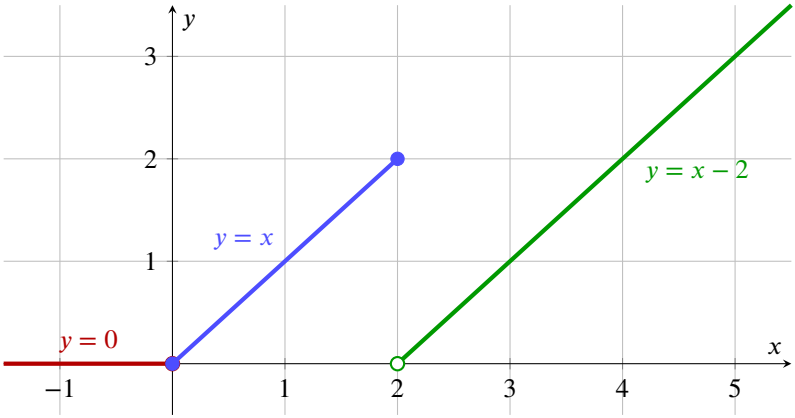
1.2 Instaneous Rate-of-Change (IRoC):

The ARoC of a function is good enough to understand the general direction in which a graph is continuing. Its accuracy can be improved by limiting the amount of change between x_1 & x_2. However a problem occurs when $\Delta x \to 0$. As cannot divide by 0. To find a functions IRoC, use the derivatives by first principles.

$$\begin{aligned}
f(x) &= 3x^2 + 2 \\
f'(x) &= \lim_{\delta x \to 0} \frac{f(x + \delta x) - f(x)}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{[3(x + \delta x)^2 + 2] - [3x^2 + 2]}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{[3x^2 + 6x\delta x + 3\delta x^2 + 2] - [3x^2 + 2]}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{6x\delta x + 3\delta x^2}{\delta x} \\
&= \lim_{\delta x \to 0} 6x + 3\delta x \\
&= 6x
\end{aligned}$$

1.3 Limits in Functions:

Specifically in Piece-wise functions there can be an issue with derivatives and their interpretations. This is mostly around when the Left and Right Hand Limits not meeting on the same point, making the functions discontinuous.



As can be seem in the graph, at the point $x = 2$ there is not derivative as the function is discontinuous. However there moving in, up until that point a derivative is present. A similar story is true at the point $x = 0$. All other points along the x-axis have a derivative that exists.

1.4 Common Derivatives:

$y = f(x)$	$\frac{dy}{dx}$
c	0
x^n	$n \cdot x^{n-1}$
e^{kx}	$k \cdot e^{kx}$
a^x	$a^x \ln(a)$
$\log_a(x)$	$\frac{1}{\ln(a) \cdot x}$
$\ln(x)$	$\frac{1}{x}$
$\sin(ax+b)$	$a\cos(ax+b)$
$\cos(ax+b)$	$-a\sin(ax+b)$
$\tan(ax+b)$	$a\sec^2(ax+b)$
$\csc(ax+b)$	$-a\csc(ax+b)\cot(ax+b)$
$\sec(ax+b)$	$a\sec(ax+b)\tan(ax+b)$
$\cot(ax+b)$	$-a\csc^2(ax+b)$
$\arcsin(ax+b)$	$\frac{a}{\sqrt{1-(ax+b)^2}}$
$\arccos(ax+b)$	$\frac{-a}{\sqrt{1-(ax+b)^2}}$
$\arctan(ax+b)$	$\frac{a}{1+(ax+b)^2}$
$\sinh(ax+b)$	$a\cosh(ax+b)$
$\cosh(ax+b)$	$a\sinh(ax+b)$
$\tanh(ax+b)$	$a\text{sech}^2(ax+b)$
$\text{csch}(ax+b)$	$-a\text{csch}(ax+b)\coth(ax+b)$
$\text{sech}(ax+b)$	$-a\text{sech}(ax+b)\tanh(ax+b)$
$\coth(ax+b)$	$-a\text{csch}^2(ax+b)$
$\text{arcsinh}(ax+b)$	$\frac{a}{\sqrt{(ax+b)^2+1}}$
$\text{arccosh}(ax+b)$	$\frac{a}{\sqrt{(ax+b)^2-1}}$
$\text{arctanh}(ax+b)$	$\frac{a}{1-(ax+b)^2}$